10.1 Information Extraction

Pull structured fields out of free text.

These notes walk through the lecture slides. The original deck is unchanged; use **(slides)** on the course hub if you want that PDF.

1. What information extraction is

Classification gives a **label** for a whole document. Information extraction (IE) pulls **fields** out of the text: who, what, when, how much, and how those pieces relate.

The input is still messy language. The output is closer to a row in a table. A news desk, a bank app, or a chatbot can act on that row.

This week follows *Practical Natural Language Processing* (Vajjala, Majumder, Gupta, Surana) and *Natural Language Processing in Action* (Lane, Howard, Hapke).

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2. Where you see it

IE is not one product. It is a family of jobs that show up inside other products.

Application	What you extract
News tagging	People, companies, places, events so a story can be filed and linked
Chatbots	The object of the request ("order a pizza") so the bot can ask the next question
Social media	Time-sensitive facts: traffic, weather, disaster updates
Forms and receipts	Amounts, dates, account numbers from a scan

A banking app that lets you photograph a check is doing IE: find the amount and the routing facts, then post the deposit.

Chatbots need the same skill. "I want to order online" is not a class label. The system has to see the **intent** (order) and then ask what to order.

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3. One paragraph, many IE jobs

A human reader of an Apple earnings story sees at once: the company is Apple Inc., Luca Maestri is the finance chief, and the event is a stock buyback.

> SAN FRANCISCO — Shortly after Apple used a new tax law last year to bring back most of the $252 billion it had held abroad, the company said it would buy back $100 billion of its stock.
>
> On Tuesday, Apple announced its plans for another major chunk of the money: It will buy back a further $75 billion in stock.
>
> "Our first priority is always looking after the business and making sure we continue to grow and invest," Luca Maestri, Apple's finance chief, said in an interview. "If there is excess cash, then obviously we want to return it to investors."
>
> Apple's record buybacks should be welcome news to shareholders, as the stock price is likely to climb. But the buybacks could also expose the company to more criticism that the tax cuts it received have mostly benefited investors and executives.

A machine has to climb a ladder of tasks:

Task	What it answers on that story
Keyphrase extraction (KPE)	The piece is about *buyback* and *stock price*
Named entity recognition (NER)	*Apple* is an organization; *Luca Maestri* is a person
Named entity disambiguation / linking	This Apple is Apple, Inc., not a fruit company or another firm with the same name
Relation extraction	Maestri **is finance chief of** Apple
Event extraction	The article is about one event: a buyback
Temporal IE	Dates and times for a calendar or a timeline

You will not build every layer this week. You need the map. Note **10.2** is the pipeline. Note **10.3** is KPE. Week 11 is NER and the tasks that sit on top of it.

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4. Why this is harder than classification

A spam filter can be wrong on one email and still be useful. An extractor that mis-labels *Apple* as a fruit, or that misses the dollar amounts, writes a **bad row**. Downstream search, a knowledge graph, or a bot will repeat that error.

IE also depends on earlier NLP steps. If sentence splitting or part-of-speech tagging is wrong, NER and relations usually follow it downhill. That is why the next note spends time on the pipeline, not only on the end task.

Start with the row you need. If the product only needs tags for a filter, stop at KPE or NER. If it needs a knowledge graph, budget for linking and relations. Do not train the whole ladder because a survey paper listed every rung.

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5. What this week is for

By the end of Week 10 you should be able to:

- name the main IE applications and the tasks in the table above
- say which task is KPE, which is NER, and which is a relation
- explain why a chatbot and a receipt scanner are both IE systems

A useful habit on every new corpus: write one paragraph, underline the spans you would want in a table, then label each underline with a task name. If you cannot name the task, you are not ready to pick a model.

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6. Practice

1. Take one product you use (mail, maps, a bank app). Name two IE tasks from the table that it is doing.
2. In the Apple snippet, which span is an entity and which fact is a **relation**?
3. Why is "the article is about a buyback" not the same job as "Luca Maestri is Apple's finance chief"?

Keep the Apple paragraph. You will reuse it in notes **10.2**, **11.1**, and **11.2**.